Introduction to the Experience Requirement for Licensure and Competency Based Assessment

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Engineering & Geoscience – Per the Act

(h) "practice of engineering" means reporting on, advising on, evaluating, designing, preparing plans and specifications for or directing the construction, technical inspection, maintenance or operation of a structure, work or process

- **If you are using your engineering knowledge and skills in your job, you are required to be licensed**

Engineering in Canada

- A self regulated profession
  - Governed by Provincial Acts
  - Requires licensure
  - Subject to discipline and compliance
  - Similar to other professions, such as Doctors, Lawyers, Architects, Nurses, etc.
- 12 Provincial /Territorial Associations
- Must register in each jurisdiction that you are practicing
- Approximately 300,000 Engineers

PEGNL

- Founded in 1952 (Geoscientists included in 1988)
- Licensing & Regulatory body for the province of NL
- Mandate – why we exist:
  - Public protection / public welfare
  - Competent practice
  - Ethical practice
- Licensure
  - Responsible for the assessment of applications for licensure; Registration Committee determine academic & experience qualifications.
- Regulation
  - Discipline & Act Enforcement
  - Code of Ethics

PEGNL

- Continued Competence
  - Mandatory Professional Development
  - Professional Development Hours (PDH"s)
  - 30 PDH's per year with two mandatory categories
  - 27 PDH's from various activities that expand or enhance your knowledge in either technical or non-technical areas.

P. Eng. Requirements

Licensure Requirements

1. Acceptable academic requirements
   - Square one is an acceptable baccalaureate degree in engineering
   - Baccalaureate degrees cover breadth of knowledge; core knowledge expected of all engineers
2. Acceptable Work Experience
3. English proficiency
   - Completion of degree in English
   - TOEFL / IELTS / ESL Courses
   - Reference responses

Licensure Requirements

4. Good character

- Plead or found guilty of any offence under statute?
- Found in any action civilly liable?
- Suspended, disqualified, censured, by any institute or organization?

5. Successful completion of the Professional Practice & Ethics Exam (PPE)

- National, computer-based exam, either written in lab or by virtual proctoring
- 65% pass with 110 multiple choice questions related to practice

Experience Requirement

- Minimum 48 months required for P. Eng. or P. Geo. licensure
- Regulations state that up to 12 months qualifying work experience prior to graduation can be credited
  - Must be at or past the midway point of the academic program
  - Document your experience, including employer, supervisors, contact info, and an overview of major responsibilities and projects

Experience Requirement

- 12 months experience credit may be given for a completed Masters or PhD in an applicable area of study
- Maximum of 24 months experience credit for pre-graduation and post-graduate studies (i.e. must gain at least 24 months practical experience)
- Canadian environment competencies experience
- International experience counts

Supervision

- By Professional Engineer or "equivalent"
  - P.E. (USA)
  - C. Eng. (UK)
  - C.PEng. (Australia)
  - Other similar designation
  - Supervisor with other qualifications – (Resume will be required; Committee must determine if qualified to judge; at least two supervisors should be a P. Eng. or equivalent)

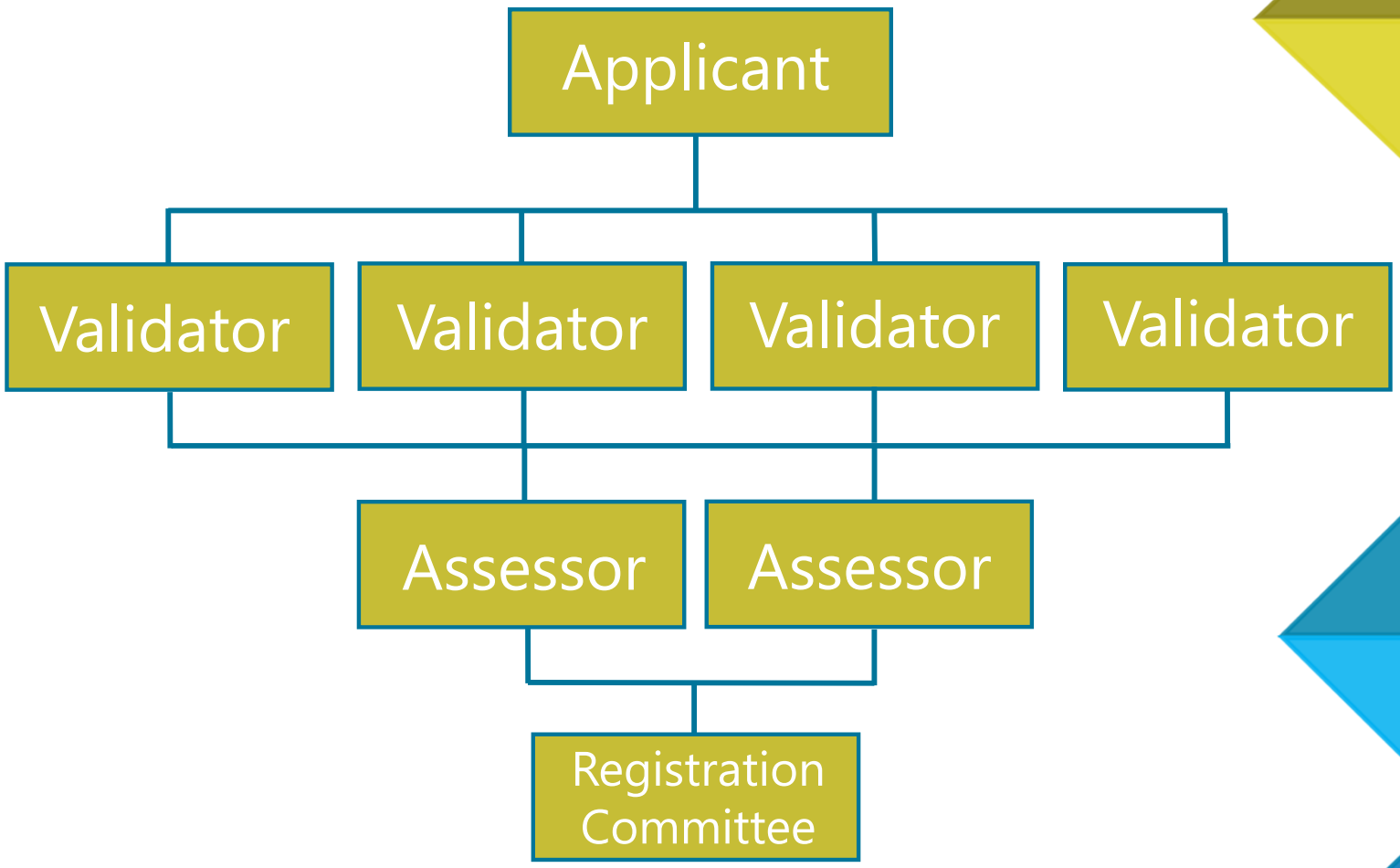
CBA

- CBA (Competency Based Assessment) provides a model for reviewing abilities and demonstrated skills and experience
- The system involves a multi-step review process
- Using an online system, applicants describe actions and outcomes which have been reviewed and rated by references known as Validators

CBA

- Third-party assessors review to determine if the competencies have been met
- This information is subsequently provided to the regulator which will use the information to help determine if the applicant is prepared for licensure

CBA


Applicant
Validator
Validator
Validator
Validator
Assessor
Assessor
Registration Committee

CBA Process - Applicant

- The first step involved in Competency Based Assessment is for the applicant to complete an online self-assessment
- The applicant will rate themselves on a scale of 0-5 in a number of competencies (34) in a handful of categories specific to their practice
- The applicant will also provide examples for each competency to support the self-evaluation and will complete an employment history with chronology and task details

CBA Process - Validator

- The validator is a reference chosen by the applicant to review the self-assessment and the ratings given by the applicant and provide comments/amendments to same
- The validators will also answer questions provided to act as a reference for the applicant and comment on their ability to independently practice
- 4 validators are required per applicant with at least 2 needing to be a Professional Engineer, Professional Geoscientist or similar

CBA Process - Assessor

- Assessors evaluate the information shared in relation to the competency examples provided by the applicant and validators
- Each file will have at least two independent assessors. This is to ensure fairness and transparency in the process
- The assessments of the applications will be shared with the Registration Committee who will use them in evaluating the overall competency of the applicant

7 Engineering Categories

- Technical Competence (10 competencies)
- Communication (3 competencies)
- Project & Financial Management (5 competencies
- Team Effectiveness (2 competencies)
- Professional Accountability (6 competencies)
- Social, Economic, Environmental & Sustainability (5 competencies)
- Personal Continuing Professional Development (3 competencies)

Competency Format

- The format which an applicant uses to demonstrate competencies is as follows:

Situation: A brief overview of a specific situation or problem

Action: The action taken in response to the situation, including engineering or geoscience judgements made or solutions found

Outcome: The impact the applicant's actions, solutions, or judgments generated

Competency Format

1.1 Regulations, Codes & Standards — **Required Overall Level: 3**

Competency to be demonstrated:
Comply with relevant legislation, regulations, and statutory reporting requirements

Workplace Examples:
1. Apply for licenses and permits
2. Undertake stakeholder consultations
3. Complete and file reports and notifications

Employer * — Select an Employer

Your Position *

Validator * — Select a Validator

Start Date * — Select Month | Select Year

End Date — Select Month | Select Year

Situation *

Save as Draft | Save as Complete | Cancel

Competency Scoring

- The reporting system uses the following scale:

0 = No Exposure

1 = Awareness/Training Level

2 = Limited Experience

3 = Approaching a Professional Level

4 = Working at a Professional Level

5 = Mature Professional Level

Documentation for CBA

- Must be an EIT or apply for licensure to access CBA system
- You can do the following now:
  - Make documentation of your experience, including employer, supervisors, contact info, and an overview of major responsibilities and projects
  - Become familiar with the Engineering Competencies and their indicators

CBA Tips

- Review the competencies regularly to make sure you are getting the experience you need to meet them
- Be honest in your self-assessment ratings
- Provide sufficient detail in your examples
- Describe what you have done, not your team
- Use PEGNL resources on website and online learning platform
- Do a quality check before submitting!
  - This is a professional document and the quality should reflect that

License vs. Internship

- P. Eng.
  - Mandatory – practice and use of title
  - Academics assessed
  - Experience assessed
  - Character assessed
  - Language assessed
  - PPE must be completed
  - Mandatory Continuing PD program requirements
  - Adherence to Code of Ethics

License vs. Internship

- E.I.T. Not mandatory – but use of title restricted
  - Not required to become an E.I.T before professional licensure
  - Academics are assessed
  - Character assessed
  - Language assessed
  - Structured Internship Program
    - Mentorship opportunities
    - Progress evaluations
    - Eligible to write the NPPE during Internship period

License vs. Internship

- E.I.T. / G.I.T.
  - Marketability
  - Networking opportunities
  - Opportunity to participate in group benefit plans

Questions?